Temporal Retiming Attacks on Hybrid Quantum Spiking Classifiers

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Abstract. Event timing is an adversarial input dimension, but attack comparisons depend on both the feasible retiming and the computation allocated to the search. We study this issue in QSNN-v3, an implemented hybrid of a classical Conv/LIF frontend and an analytically simulated eight-qubit probability readout. The model achieves $97.272\% \pm 0.212\%$ validation accuracy across five seeds conditional on shared pretraining and split. On 100 jointly clean-correct seed-42 validation examples, timestamp PGD succeeds on three examples at each tested radius. Derivative-free margin search produces zero to two additional successes depending on candidate, forward-pass or elapsed-time matching; no paired comparison establishes significant superiority. Separate classical packet-grid controls show that similar clean accuracy can accompany sharply different outcomes under one attack recipe, without isolating a representation or quantum effect. The evidence supports protocol-specific temporal vulnerability; official-test hybrid robustness and a quantum-specific mechanism remain unmeasured.

Keywords: event-driven classification; adversarial retiming; hybrid quantum classifiers; spiking neural networks; matched attack evaluation.

1 Introduction

Event-driven classifiers process not only where an event occurs but when it occurs. Retiming can change a prediction while preserving event identities, spatial coordinates, polarity and total count. Surrogate-gradient learning makes spiking dynamics trainable [1], while spike-retiming attacks make timing an explicit adversarial input dimension [2]. An evaluation must therefore specify both the admissible temporal changes and the computation available to find them.

We investigate end-to-end temporal vulnerability in an implemented hybrid classifier, emphasizing how computational matching changes attack comparisons and why results from separate temporal representations require separate interpretation. Our primary target, QSNN-v3, maps events through a classical Conv/LIF frontend, learned angles, an eight-qubit circuit and an affine probability readout. The central question is whether conclusions about its observed attack outcomes change with the definition of matched computation. Candidate count, forward-pass count and elapsed time measure different resources, particularly when only one attack uses gradients.

On the same 100 joint clean-correct validation examples, timestamp PGD causes three hybrid failures at each tested radius. TEMP-DRIFT-v2, a derivative-free margin search, causes 3–5 depending on radius and computational control. The largest numerical difference occurs under elapsed-time matching; no paired test establishes superiority. A separate four-model classical packet-grid study supplies context: similar clean accuracy need not imply similar attack outcomes. Differences in victims, populations, checkpoints and budget units prevent a causal representation ablation or a pooled robustness ranking across these studies.

The evaluation concerns the implemented composition. Quantum-classifier robustness can depend on encoding, model structure and adversarial access [3–5]; here the perturbation enters before the classical frontend. The following concrete outputs organize the paper:

1. **An explicit hybrid attack interface.** We specify temporal aggregation, Conv/LIF features and the sole quantum-probability readout, with preserved implementation evidence and a readout-specific margin observation. Missing artifacts limit full reproduction.
2. **Paired computational comparisons.** We report success counts, overlaps, exact paired tests and resource accounting for candidate, forward and elapsed-time controls on a frozen timestamp population.
3. **Classical interpretation controls.** We compare four architectures under a common packet-retiming recipe to delimit what clean accuracy and one optimizer establish.

Source labels [R1]–[R10] map to preserved repository reports and code in the supplement. The supplement also retains development studies, full numerical tables and the conceptual literature comparison.

2 Related work

Raw-event attacks and constrained retiming. Surrogate-gradient learning provides a practical route through the non-differentiable dynamics of spiking networks [1]. Büchel et al. study event additions and removals in spiking vision models [6], a different feasible set from identity-preserving retiming. An attack on an aggregated tensor and an attack on the underlying event stream need not have the same feasible set. Yao et al. attack raw event data using a probabilistic treatment of discrete event values and sparsity control [7]. Their construction permits event-value and support changes, whereas our native-event threat model preserves event identities, coordinates, polarities and count. Du et al. investigate raw-event attacks with configurable latencies [8], providing a particularly relevant connection to temporal manipulation. These works motivate attacks at the event interface without making their constraints interchangeable with ours.

Capacity constraints and representation-specific budgets. Yu et al. formalize capacity-one spike retiming and budgets on maximum displacement, total delay and the number of retimed spikes [2]. Our classical packet-grid comparison uses a local implementation of this framework, with completed results for maximum displacement and retimed-packet count. The native-timestamp experiment instead permits multiple events to contribute to a frame cell. Its duration-scaled timestamp radius cannot be identified with a one-bin packet displacement. We consequently compare attacks within each stated regime and do not rank the regimes by their raw ASR values.

Hybrid spiking and quantum models. Quantum temporal models encompass distinct computational constructions. Brand and Petruccione develop a quantum leaky integrate-and-fire neuron and network [9]; Jha et al. combine SNN-derived EEG features with quantum-kernel classification [10]; and SPATE studies spike-derived temporal quantum encodings [11]. QSNN-v3 instead composes classical Conv/LIF dynamics, a learned angle map, a data-re-uploading variational circuit [12], and a linear classifier on the complete measurement-probability vector. These differences concern the model and interface, so accuracy or security claims do not transfer directly between the constructions.

Adversarial robustness of quantum classifiers. Wendlinger et al. compare quantum and classical classifiers using transfer attacks, perturbation structure and Lipschitz-based analysis [3]. Gong et al. investigate randomized encodings under specified adversarial-access assumptions [4], while Dowling et al. establish robustness guarantees for particular quantum-classifier settings [5]. Khatun and Usman use classical autoencoders to distill quantum adversarial manipulations on image benchmarks [13]; this is neither an implemented defense here nor evidence for temporal-retiming protection. These studies motivate stating the encoding and adversary explicitly. Their guarantees and defenses cannot be assigned to QSNN-v3 without checking the corresponding assumptions, especially because our perturbation enters before a classical event-processing frontend.

Evaluation strength and computational controls. PGD is a standard iterative baseline [14], but the failure of one optimizer is insufficient evidence of robustness [15, 16]. Diverse attacks and improved optimization can expose failures missed by an individual configuration [17]. We therefore report paired outcomes under candidate, forward-pass and elapsed-time controls. These controls answer different resource questions and do not establish optimality of either attack. Standard image-space attack ensembles also require adaptation before they respect our order, timing and packet constraints.

The supplementary conceptual-positioning table distinguishes these interfaces without ranking external ASRs.

3 Models and representations

Figure 1 traces the attacked timestamp-to-logit composition; Table 2 separates the experimental regimes.

3.1 Native events and temporal-grid packets

The timestamp study uses native N-MNIST events $e_i = (x_i, y_i, p_i, t_i)$. Event identities, coordinates, polarity, labels and event count are immutable. Timestamps are assigned to ten equal-duration bins using the clean recording endpoints. The differentiable adapter has hard count accumulation in the forward pass and triangular interpolation weights for the backward pass. Counts are capped at 255 by the adapter. The hybrid frontend then applies $\min(X/8, 1)$. These input transformations are model preprocessing; the preservation claim refers to the underlying events.

Table 1. Trainable parameters of QSNN-v3.

Component	Parameters
Conv/LIF frontend and 32D summary	37,792
Learned $32 \to 8$ projection	264
Angle map and quantum head	136
Probability-to-logit classifier	2,570
Total	**40,762**

The quantum head includes the learned $8 \to 8$ angle map, two upload blocks and learned measurement basis. Nontrainable BatchNorm buffers and state-vector storage are excluded [R1].

The grid study instead freezes a ten-bin equal-event-count representation. Each nonzero cell on a fixed spatial/polarity line is one amplitude-bearing packet. A binary packet has value one; an integer packet carries its complete cell count. A packet with count five remains a single packet carrying five during retiming. Collision-free capacity-one projection prevents packet splitting, merging, deletion, creation and amplitude changes. This distinction matters because multiple raw events can occupy one cell in the timestamp study, while one grid packet is indivisible in the second study.

3.2 The implemented hybrid classifier

The Conv/LIF extractor uses two 3×3 bias-free convolutions with 16 and 32 output channels, batch normalization and average pooling. LIF dynamics use decay 0.5. Time-averaged spike activations and the final membrane state are pooled to 4×4 spatial summaries and concatenated. A learned projection followed by tanh produces 32 latent features. A learned 32-to-8 projection and another tanh feed the circuit head. An additional learned 8-to-8 projection, initialized to the identity, produces bounded rotation angles.

For the latent vector $h \in \mathbb{R}^{32}$, the learned angle map is

$$\begin{aligned} u &= \tanh(W_{32}h + b_{32}), \\ a &= \pi \,\mathrm{clip}(W_8 u + b_8, -1, 1). \end{aligned} \tag{1}$$

The eight-qubit register starts in $|0\rangle^{\otimes 8}$. Each of two blocks applies an RY upload of each angle, an RZ upload scaled by a learned phase parameter, learned RY/RZ gates, and a ring of CNOT gates. Learned local RZ/RY rotations set the measurement basis. The model returns all 256 computational-basis probabilities to a linear ten-class classifier. All paths from the latent features to the logits pass through the circuit; there is no direct latent-to-logit branch.

$$\begin{aligned} p_k &= \left|\langle k|U_\theta(a)|0\rangle^{\otimes 8}\right|^2, \quad k = 0, \ldots, 255, \\ z &= W_{\mathrm{out}}p + b_{\mathrm{out}}. \end{aligned} \tag{2}$$

The backend is an analytic batched PyTorch state-vector simulation with complex64 amplitudes. Probabilities are computed exactly within numerical precision. The experiments do not involve quantum hardware, finite-shot readout or hardware noise. The name QSNN-v3 identifies this hybrid implementation and should not be interpreted as a network of physical quantum LIF neurons [R1].

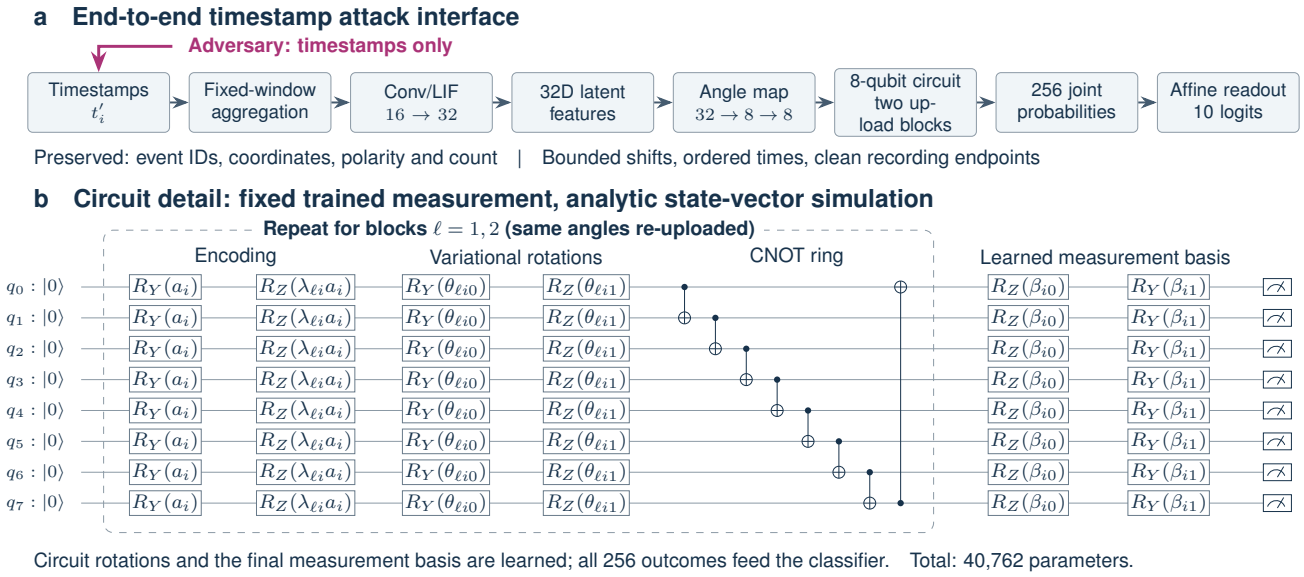


Figure 1. Timestamp attack interface and QSNN-v3 computation. Only timestamps are mutable; the aggregation window is fixed by the clean recording. Each of two circuit blocks re-uploads the angles, applies variational rotations and a CNOT ring; a learned measurement basis precedes the 256-probability affine readout. The diagram specifies computation, not an observed attack example. Source: [R1–R3].

3.3 Readout-specific margin observation

For the fixed trained affine readout $z(q) = Wq + b$, let q, q' be clean and retimed measurement probabilities and $m_{yk}(q) = z_y(q) - z_k(q)$ for the clean predicted class y. With states ρ, ρ' before the fixed measurement and $D_{\mathrm{tr}} = \frac{1}{2}\|\rho - \rho'\|_1$, measurement contraction [18] and Hölder's inequality imply

$$\begin{aligned} m_{yk}(q') &\geq m_{yk}(q) \\ &- 2\|w_y - w_k\|_\infty D_{\mathrm{tr}}(\rho, \rho'). \end{aligned} \tag{3}$$

Here w_k is row k of W and the state norm is the trace norm. Strict positivity of the right-hand side for every $k \neq y$ suffices to preserve the label. The supplement gives the derivation. No uniform retiming-to-state bound or final-model per-example state measurements are available, so this is an analytical interpretation, not a certified timestamp radius.

4 Threat models and attacks

4.1 Timestamp retiming

We consider untargeted test-time evasion. For a sorted clean timestamp vector t, the feasible retiming keeps each event within its clean timestamp neighborhood and within the recording endpoints, and preserves a nondecreasing sequence. If D is the clean duration, the perturbation budget is $\varepsilon = \varepsilon_f$ D. The reported matched comparisons use $\varepsilon_f = 0.05$ and 0.10. Equal timestamps are permitted; capacity-one grid occupancy is not part of this threat model.

$$\begin{aligned} &|t'_i - t_i| \leq \varepsilon_f D, \qquad t_1 \leq t'_i \leq t_N, \\ &t'_i \leq t'_{i+1}, \qquad D = t_N - t_1 + 1. \end{aligned} \tag{4}$$

The extra unit in D follows the implementation's inclusive timestamp convention. The implementation first clips a candidate to eventwise intervals, applies a cumulative maximum, and clips to the recording range. We describe this as a feasibility operator, not as the Euclidean projection onto the constrained set. Event counts over the fixed recording window are preserved. Inter-spike intervals, time-bin counts and detector statistics can change. No completed final-model experiment establishes statistical indistinguishability from clean inputs.

Timestamp PGD. Timestamp PGD uses white-box gradients of cross-entropy through the hard-forward/soft-backward frame adapter and the surrogate-gradient LIF model. It starts at the clean timestamps, takes 20 sign-gradient steps with step size $\varepsilon/5$, and applies the feasibility operator after every step. The candidate with the highest cross-entropy among the clean input and 20 iterates is retained. There is no random start in the preserved implementation [R2]. Selection does not explicitly retain any successful intermediate candidate: a higher-cross-entropy candidate may still be correctly classified. This is a verified selection rule, not evidence that a reported success was lost; testing its empirical effect requires the missing trajectories.

$$\begin{aligned} g^{(k)} &= \operatorname{sign}\left(\nabla_t \mathrm{CE}(z(t^{(k)}), y)\right), \\ t^{(k+1)} &= \Pi_\varepsilon\left[t^{(k)} + \frac{\varepsilon}{5} g^{(k)}\right]. \end{aligned} \tag{5}$$

TEMP-DRIFT-v2. TEMP-DRIFT-v2 is a derivative-free search using model logits and the true label. It minimizes the true-class margin. A negative margin indicates that a competing logit exceeds the correct-class logit. This objective is a classification objective and contains no fidelity, density-matrix distance or classical-statistics penalty.

$$\min_{t' \in \mathcal{F}_\varepsilon} m(t'), \qquad m(t) = z_y(t) - \max_{j \neq y} z_j(t). \tag{6}$$

The preserved full-budget implementation evaluates 600 uniformly perturbed feasible candidates. It retains the 12 candidates with lowest margin, then runs four generations of 250 children. Each child combines an elite parent, half the difference between two elite candidates and Gaussian

noise at scales 0.50, 0.30, 0.18 and 0.10 times ε. Every proposal is made feasible before evaluation. The final candidate with lowest margin is returned. This configuration uses 1,600 candidate evaluations [R2]. The matched implementation uses 13 initial candidates plus four generations of two children for $Q = 21$, and 13 plus four generations of seven for $Q = 41$. It always keeps 12 elites; the four noise scales are unchanged. For elapsed-time budgets it sets $n_0 = \max(12, \mathrm{round}(0.375Q))$ and $n_g = \mathrm{round}((Q - n_0)/4)$, recording the actual total $n_0 + 4n_g$. The selected $Q = 4{,}000$ and 1,600 give $1{,}500 + 4 \times 625$ and $600 + 4 \times 250$, respectively. The supplement gives the complete schedule and pseudocode [R3].

TEMP-DRIFT-v2 is the historical implementation name; throughout the final timestamp comparison it denotes margin search.

Inference state. Both timestamp victims are loaded in evaluation mode, fixing BatchNorm statistics. Their forward methods initialize membrane and output accumulators locally on every call; the preserved code has no cross-candidate persistent neuron state. This static inspection covers the timestamp models, not the missing four-model grid implementations.

4.2 Discrete packet retiming

In the second study, let s_i and s'_i denote a source packet bin and its destination. Budgets count packet displacements independently of amplitude. A packet carrying amplitude five and shifted by two bins contributes two to B_1 and one to B_0. Capacity-one projection preserves packet identity, amplitude, event line and total represented mass.

$$B_\infty = \max_i |s'_i - s_i|, \quad B_1 = \sum_i |s'_i - s_i|, \\ B_0 = \sum_i \mathbf{1}[s'_i \neq s_i]. \tag{7}$$

The local projected-in-the-loop attack uses differentiable shift probabilities and evaluates feasible hard schedules. Only $B_\infty = 1$ and $B_0 = 200$ are complete in the controlled four-model campaign [R4]. The comparison report attributes a B_0 penalty coefficient of five to the implementation, whereas the reference paper states ten; the campaign override source is absent from the accessible snapshot. This discrepancy limits exact-replication claims. Historical sweeps from other checkpoints appear in the supplement.

5 Experimental protocol

5.1 Training and model selection

N-MNIST contains event recordings of handwritten digits [19]. The hybrid campaign uses a frozen stratified partition of 55,000 training and 5,000 validation examples from the official training set. Architecture selection was performed with seed 42 and validation data. The chosen wide4 frontend and project_measure2 circuit were then held fixed across model seeds 42, 123, 777, 2026 and 6543. The official test partition was not instantiated for the final QSNN campaign [R1]. Consequently, its reported clean accuracy is validation accuracy after model selection.

All five hybrid runs load the same hash-verified, seed-42 classical frontend checkpoint. The frontend is frozen for the first two epochs, then optimized with its BatchNorm layers kept in evaluation mode. The random seeds vary subsequent initialization and training stochasticity; they do not represent five independently pretrained frontends or five data splits. This conditional design limits the scope of the reported between-seed standard deviation.

Training uses AdamW with weight decay 10^{-4} and batch size 256 for at most 40 epochs. Learning rates are 5×10^{-5} for the extractor, 2.5×10^{-4} for the projection/circuit, and 10^{-3} for the classifier. A ReduceLROnPlateau scheduler uses factor 0.5, patience 2 and minimum rate 10^{-5}. The highest validation accuracy selects a checkpoint, with validation loss breaking ties; early stopping uses patience 6 after at least eight epochs. Gradients are clipped at norm 1. The frontend uses float16 autocast; circuit inputs and amplitudes use float32 and complex64. Seed 42 was externally interrupted during epoch 38; its saved epoch-37 checkpoint was re-evaluated [R1].

5.2 Populations and attack budgets

The timestamp protocol freezes 100 validation examples, with ten examples per class, correctly classified by both models before attack. It reuses their identities across conditions. This joint-correct restriction permits paired attack comparisons but excludes examples that only one model classifies correctly. It is not a population-wide robust-accuracy estimate. The compact SNN checkpoint in this study differs from the retrained custom-SNN checkpoint in the controlled grid study.

PGD evaluates 21 candidate timestamps, including the clean candidate. Its implementation makes 41 attack-internal forward passes and 20 backward evaluations because candidate scoring and gradient computation use separate forwards. TEMP uses Q forward evaluations and no backwards. The first matched condition gives TEMP 21 candidates; the second gives it 41 forwards. Neither converts a backward pass into an assumed number of forward passes. Verification adds two forwards per new record and is excluded from attack timing.

Elapsed-time calibration selects Q = 4,000 for QSNN and Q = 1,600 for SNN. The reported median TEMP/PGD timing ratios are 0.978 and 1.083, respectively [R3]. Candidate matching, forward matching and elapsed-time matching answer different computational questions. Exact equality of elapsed time is not asserted, and no throughput or latency claim is inferred from the ratios alone. The source synchronizes CUDA around timing, warms up PGD and TEMP, and uses ten disjoint training calibration examples (one per class), two repeats and $\varepsilon_f = 0.10$. TEMP evaluates candidate chunks of at most 64; PGD handles one example per forward. Timestamp tensors are float32 and circuit amplitudes complex64; training autocast settings are not assumed to describe attack timing. Saved timing arrays and a run-specific hardware/precision snapshot are unavailable. The supplement records calibration details and remaining provenance limits.

The controlled grid campaign trains four architectures under a common local recipe, data split, seed, optimizer settings and validation-selected checkpoint rule [R4]. For each representation it selects the first 1,000 official-test examples that all four models classify correctly. The joint-correct pools contain 9,522 binary and 9,495 integer examples. Within each

representation the same frozen manifest is used for every architecture and budget. The two representations use different manifests. Exact baseline definitions, training configs and checkpoint files are absent from this snapshot; architecture names alone do not verify a particular published implementation. We therefore retain the report labels without assigning an unverified architecture citation.

5.3 Metrics, paired tests and evidence

Clean accuracy uses the complete stated partition. ASR is the fraction of the stated clean-correct attack population that becomes misclassified. Thus $1-\mathrm{ASR}$ is survival under the evaluated search on that selected subset, not unconditional robust accuracy. Joint-correct selection improves within-study comparability but changes the estimand relative to all examples.

For the stated clean-correct attack population $\mathscr{P}$, we use

$$\mathrm{ASR}(\mathscr{P}) = \frac{1}{|\mathscr{P}|} \sum_{i \in \mathscr{P}} \mathbf{1}\left[f(X_i') \neq y_i\right]. \tag{8}$$

Figure 2 uses 95% Wilson score intervals computed from the reported numerator and $n = 100$; the formula is given in the supplement. These are descriptive binomial-model intervals, with no sampling-based claim for the deterministically selected population or across reused conditions. They are not intervals for the paired difference. We report arithmetic means and sample standard deviations across the stated model seeds. For paired timestamp comparisons, b counts examples broken only by PGD and c counts examples broken only by TEMP. The exact two-sided McNemar test uses a Binomial$(b+c, 1/2)$ null distribution, with $p = 1$ for $b+c = 0$. It assesses observed paired discordance, not equality of all possible attack strengths. Reused examples across seeds and conditions are not treated as independent additional subjects.

All numerical results are taken from preserved reports and source files at repository snapshot `80d02f5733d4` [20]; the full commit identifier is recorded in the supplement. We checked parameter arithmetic, aggregate statistics, paired counts and exact paired p-values. Model execution and per-example bootstrap resampling were not repeated. Missing raw outputs and provenance boundaries are documented in Sec. 7 and the supplement.

6 Results

The primary result is the paired timestamp comparison on QSNN-v3. Clean validation establishes the victim's predictive performance; separate packet-grid controls delimit its interpretation. Full seed, overlap and grid tables are retained in the supplement.

6.1 Clean validation of the hybrid classifier

QSNN-v3 reaches 97.2720% ± 0.2119% validation accuracy and reported macro-F1 of 97.2736% ± 0.2119% across five training seeds [R1]. The sample SD is conditional on shared pretraining and split, as specified in Sec. 5; it is not attack-rate uncertainty. The preserved SNN validation reference reaches 98.1320% ± 0.2407% accuracy, a descriptive QSNN-minus-SNN difference of -0.8600 percentage points. A matched validation-F1 comparison is unavailable.

The seed-42 wide4 classical frontend reaches 97.80% validation accuracy with 38,122 parameters, compared with 97.06% and 40,762 parameters for its hybrid continuation. Different training stages and budgets prevent attributing this difference to an isolated circuit component. The supplement retains individual accuracies, selected epochs and development records; no quantum accuracy advantage or official-test hybrid result is established.

6.2 Matched timestamp attacks

Each cell uses the same 100 joint clean-correct validation examples (seed 42), so one success equals one ASR percentage point. Figure 2 groups the candidate and forward conditions because their aggregate outcomes coincide; these conditions remain different resource definitions. At both radii, PGD succeeds on 3/100 examples. TEMP succeeds on 4/100 and 3/100 at 5% and 10% under either access control, versus 4/100 and 5/100 under elapsed-time calibration. The latter uses $Q = 4{,}000$ for QSNN-v3, with a reported median TEMP/PGD runtime ratio of 0.978 [R3].

The paired comparison is determined by discordant examples. At 5%, all controls have three shared successes, one TEMP-only success and 96 failures of both attacks ($p = 1$). At 10%, access matching has three shared successes and 97 failures of both ($p = 1$); elapsed-time matching has three shared, two TEMP-only and 95 failures of both ($p = 0.5$). There are no PGD-only successes in these cells. Thus two examples account for the largest numerical contrast; no matched condition establishes TEMP superiority, and nonsignificance does not prove equivalence.

For the compact SNN, candidate and forward matching produce no successes at either radius. Elapsed-time matching uses $Q = 1{,}600$ and a median runtime ratio of 1.083, producing 0/100 versus 0/100 at 5% and 0/100 PGD versus 1/100 TEMP at 10% ($p = 1$). These selected-population observations do not estimate full-set robust accuracy or general hybrid-versus-SNN robustness. The supplement preserves every matching condition, reported bootstrap differences and attributed distortion summaries.

6.3 Separate packet-grid controls

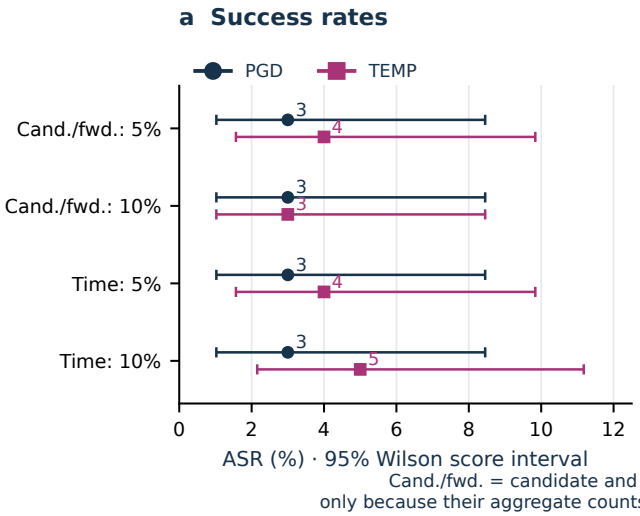
The grid study uses different victims, capacity-one packets and its own 1,000-example joint clean-correct test manifest for each representation (Sec. 5). Under the two completed budgets, custom-SNN ASR spans 98.0–100%, whereas the three reference architectures span 0–1.9% (Fig. 3). Clean test accuracy, measured on all 10,000 examples, is shown separately in Table 3; the supplement supplies all sixteen attack cells.

The integer custom SNN and spiking ResNet18 differ by only 0.02 percentage points in clean accuracy, yet by 99.7 and 97.9 ASR points under $B_\infty = 1$ and $B_0 = 200$. Clean accuracy therefore does not summarize the failure profile under this recipe. Common settings do not guarantee equally strong optimization across architectures, and zero observed successes are not robustness certificates [16, 17]. These are

Table 2. Experimental populations and budget units. These regimes are analyzed separately.

Study	Models	Attack population	Budget
Native timestamps	QSNN-v3 and compact SNN	100 common clean-correct validation examples; 10/class; seed 42	$\varepsilon_f \in \{0.05, 0.10\}$ of recording duration
Controlled grid	Custom SNN, ConvNet, spiking ResNet18, VGGSNN	First 1,000 common clean-correct official-test examples per representation; seed 42	$B_\infty = 1$ bin or $B_0 = 200$ packets
Supporting studies	Earlier models/checkpoints	Study-specific subsets stated in the supplement	Study-specific timing or packet budgets

The compact timestamp SNN and retrained grid custom SNN are different checkpoints. Binary and integer grid attacks use separate manifests, drawn from joint-correct pools of 9,522 and 9,495 test examples, respectively [R2–R5].



b Paired counts (n=100 per row)

Both	PGD-only	TEMP-only	Neither
3	0	1	96
3	0	0	97
3	0	1	96
3	0	2	95

Figure 2. QSNN-v3 timestamp outcomes on the same $n = 100$ joint clean-correct validation examples, seed 42. (a) ASR with descriptive 95% Wilson score intervals; numbers beside markers are success counts. Candidate ($Q = 21$) and forward ($Q = 41$) matching share the displayed aggregate counts. (b) Exact paired counts, including the small discordant groups. Equal aggregate rows for candidate and forward matching do not establish the same failed IDs across these conditions. Intervals are binomial-model summaries for this selected subset, not independent evidence or uncertainty over all validation/test examples. Source: [R3].

Table 3. Clean accuracy (%) on all 10,000 official-test examples. ASR uses separate subsets.

Architecture	Binary	Integer
Custom SNN	97.99	98.14
ConvNet	96.90	96.58
Spiking ResNet18	98.61	98.12
VGGSNN	99.21	99.30

single-seed local controls, not values copied from external benchmarks or a representation ablation.

The grid report attributes passing reconstruction/prediction audits to all cells, but the underlying artifacts are absent from the supplied snapshot. The unresolved B_0 penalty discrepancy described in Sec. 4 further limits replication claims [R4].

7 Discussion and limitations

The primary timestamp experiment identifies feasible retimings that change QSNN-v3 predictions. Its numerical PGD/TEMP contrast changes with computational matching: the two extra failures at the larger radius occur under elapsed-time calibration and disappear under access matching. The paired tests remain inconclusive. The classical controls answer a separate question: similar clean accuracy need not imply similar outcomes under the same nominal attack recipe. Different victims, samples and budget units prevent pooling the two studies.

Mechanism and encoding. The quantum circuit is one stage of an attacked composition. Timestamp changes can affect bin counts, spiking summaries, angles and measurement probabilities before the affine readout. Without fixed-frontend classical-head and circuit ablations, failures cannot be attributed uniquely to quantum processing. The margin observation explains why state displacement alone need not change a label, but final-model per-example states and margins are missing. Historical Iris diagnostics in the supplement do not supply that evidence. Encoding- and access-specific quantum analyses [3–5] consequently provide context rather than guarantees for our temporal feasible set.

Search strength. Low or zero ASR can reflect optimizer limitations. PGD uses one start and 20 steps, with cross-entropy-based output selection; TEMP uses a finite margin search. The large classical architecture gap requires gradient, preprocessing and search-strength diagnostics before interpretation as an intrinsic robustness difference. Alternative surrogates, restarts and derivative-free budget curves should be

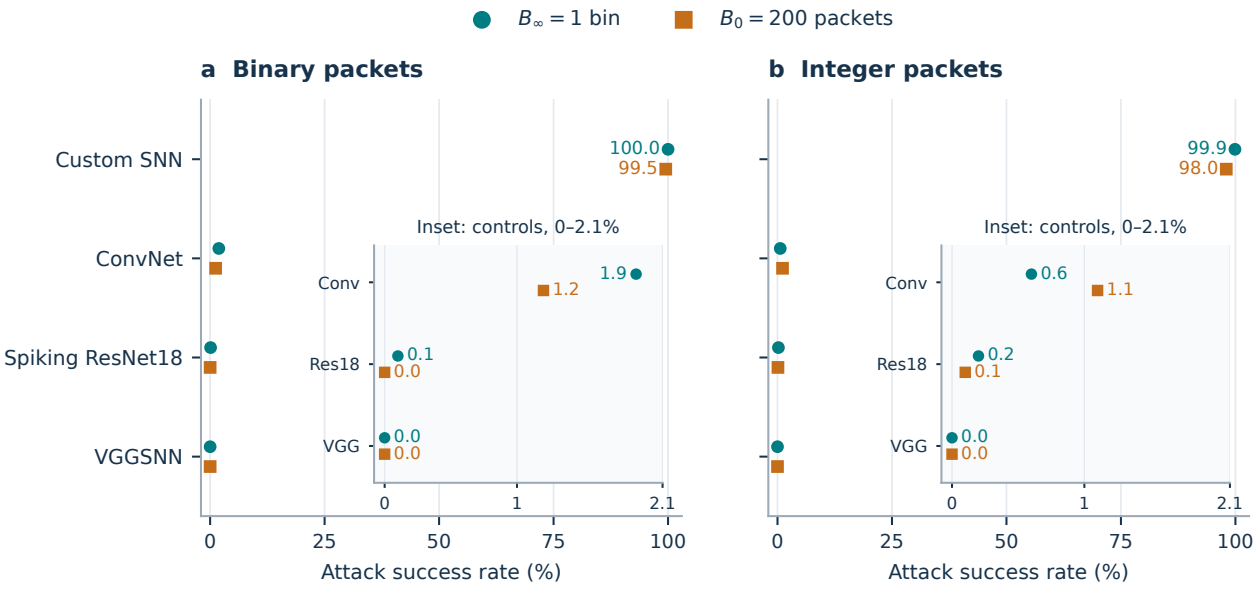


Figure 3. Separate classical controls under maximum displacement ($B_\infty = 1$ bin) and retimed-packet count ($B_0 = 200$). Each cell uses $n = 1{,}000$ joint clean-correct test examples, seed 42. Insets explicitly expand 0–2.1% for the three low-ASR controls. Binary and integer panels use different manifests; the panels are not paired representation comparisons. Source: [R4].

checked with valid hard-forward candidates [16, 17]. Image-space attack ensembles and defenses, including classical distillation of quantum manipulations [13], require adaptation to event constraints.

Evaluation and evidence boundary. Five hybrid seeds share pretraining and split, while attacks use one victim seed and a selected validation population. Official-test QSNN robustness, independently pretrained pipelines and a larger attack population remain unmeasured. The populated comparison report conflicts with its README's statement that results are absent; we use the report and disclose that inconsistency. Most checkpoints, raw arrays, manifests and logs referenced by reports are missing. Our arithmetic and static code checks do not reproduce model runs or independently repeat their attributed audits. The grid penalty ambiguity and incomplete baseline provenance remain unresolved.

Next empirical steps. After validation-based model selection, evaluate QSNN clean and attacked performance on the official test set across independently trained frontend/head seeds, retaining sample IDs and clean errors. Sweep steps, step size, restarts, cross-entropy/margin losses and surrogate choices, recording the best valid success by budget and reconfirming it with hard forwards. Hold frontend, split and training budget fixed for capacity-matched classical heads and circuit variants with/without entanglement, including parameter counts and training cost. To test the state/margin interpretation, retain per-example states, probabilities and margins and control clean margin and realized distortion; mean fidelity alone would not establish a mechanism.

A future monitor must specify eight-qubit observables, shot budget, clean-only calibration and response policy before measuring ROC/PR, false-positive rates and adaptive attacks. Single-qubit marginals do not determine the full state. Event-count preservation is also distinct from stealth: no completed statistical-indistinguishability or final-model detection/defense evaluation exists. The historical corrected grid campaign remains only 8 of 33 conditions complete.

8 Conclusion

Computational matching changes the numerical attack comparison on the implemented QSNN-v3, but none of the paired conditions establishes TEMP superiority. Constrained retiming changes 3–5 of 100 examined joint-correct validation predictions. Separate classical controls show why clean accuracy and a common recipe are insufficient summaries of attack behavior. The resulting evidence is specific to the stated protocols; independent official-test hybrid evaluation is the principal missing validation.

Data and code availability

The `ABC_def1` repository [20] and supplied snapshot contain implementation excerpts and preserved reports. The supplement maps source paths and the pinned commit. Tables transcribe those reports; figures reorganize their documented counts. Full reproduction requires omitted artifacts and original datasets. Author details remain withheld for review.

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